

ATOMIC STRUCTURE

EAPCET Previous-Year Questions (PYQs)

30 single-correct questions selected from Telangana engineering entrance examinations, 2023–2025.

30	3	1 mark
QUESTIONS	EXAM YEARS	EACH

Instructions

Choose the single best answer. Use $h = 6.626 \times 10^{-34} \text{ J s}$, $c = 3.0 \times 10^8 \text{ m s}^{-1}$, and $1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$ where required.

Source note: Questions are transcribed from the chapter-wise TS/TG EAMCET-EAPCET archive. Exam labels are retained with every question.

QUESTION 01 · TG EAPCET 2025 (Online) 4th May Evening Shift

Observe the following statements.

Statement -I Rutherford model of an atom cannot explain the stability of an atom.

Statement-II The wavelength of X-rays is higher than the wavelength of microwaves.

The correct answer is

- A Both Statements I and II are correct.
- B Both Statements I and II are not correct.
- C Statement I is correct, but Statement II is not correct.
- D Statement I is not correct, but Statement II is correct.

QUESTION 02 · TG EAPCET 2025 (Online) 4th May Evening Shift

In hydrogen atom, an electron is transferred from an orbit of radius 1.3225 nm to another orbit of radius 0.2116 nm . What is the energy (in J) of emitted radiation?

- A $1.635 \times 10^{(-18)}$
- B $3.027 \times 10^{(-19)}$
- C $4.087 \times 10^{(-19)}$
- D $0.4578 \times 10^{(-18)}$

QUESTION 03 · TG EAPCET 2025 (Online) 4th May Morning Shift

The radius of second orbit of hydrogen atom is same as that of orbit (n) of an ion (x), n and x are respectively.

- A $4, \text{Be}^{(2+)}$
- B $3, \text{Li}^{(2+)}$
- C $4, \text{Be}^{(3+)}$
- D $2, \text{He}^{(+)}$

QUESTION 04 · TG EAPCET 2025 (Online) 4th May Morning Shift

An electromagnetic radiation of wavelength 331.5 nm is made to strike the surface of a metal. Electrons are emitted with a kinetic energy of $12 \times 10^{(5)} \text{Jmol}^{(-1)}$. The work function (in eV) of the metal is ($h=6.63 \times 10^{(-34)} \text{Js}$, $N_{\text{A}}=6 \times 10^{(23)} \text{mol}^{(-1)}$)

- A 1.5
- B 3.0
- C 3.5
- D 2.5

QUESTION 05 · TG EAPCET 2025 (Online) 3rd May Evening Shift

In the atomic spectrum of hydrogen, the wavelengths of the spectral lines corresponding to electronic transitions (i) $n=4$ to $n=2$ and (ii) $n=3$ to $n=1$ are λ_1 and λ_2 Å respectively. The value of $(\lambda_1 - \lambda_2)$ (in cm) is (R_H = Rydberg constant)

- A $\frac{1}{R_H} \left[\frac{24}{101} \right]$
- B $R_H \left[\frac{24}{101} \right]$
- C $\frac{1}{R_H} \left[\frac{101}{24} \right]$
- D $R_H \left[\frac{101}{24} \right]$

QUESTION 06 · TG EAPCET 2025 (Online) 3rd May Evening Shift

Work functions of four metals M_1, M_2, M_3 and M_4 are 4.8, 4.3, 4.75 and 3.75 eV respectively. The metals which do not show photoelectric effect when light of wavelength 310 nm falls on the metals are

- A M_1, M_2 only
- B M_1, M_3 only
- C M_1, M_2, M_3 only
- D M_1, M_2, M_4 only

QUESTION 07 · TG EAPCET 2025 (Online) 3rd May Morning Shift

The energy associated with electron in first orbit of hydrogen atom is -2.18×10^{-18} J. The frequency of the light required (in Hz) to excite the electron to fifth orbit is ($h = 6.6 \times 10^{-34}$ Js)

- A 2.17×10^{16}
- B 3.17×10^{14}
- C 2.17×10^{15}
- D 3.17×10^{15}

QUESTION 08 · TG EAPCET 2025 (Online) 3rd May Morning Shift

In $Sr(Z=38)$, the number of electrons with $l=0$ is x , number of electrons with $l=2$ is y . $(x-y)$ is equal to (l = Azimuthal quantum number)

- A 0
- B 8
- C -2
- D 2

QUESTION 09 · TG EAPCET 2025 (Online) 2nd May Evening Shift

The electron in hydrogen atom undergoes transition from higher orbits to an orbit of radius 476.1 pm . This transition corresponds to which of the following series?

- A Lyman
- B Paschen
- C Balmer
- D Pfund

QUESTION 10 · TG EAPCET 2025 (Online) 2nd May Evening Shift

Identify the incorrect statement from the following?

- A m , designates the orientation of the orbital.
- B The probability density of electron is expressed by $|\psi|^2$.
- C The total information about electron in atom is stored in its ψ .
- D Total number of orbitals in a sub level is equal to $(2l+1)$.

QUESTION 11 · TG EAPCET 2025 (Online) 2nd May Morning Shift

The radius of stationary state ($n=2$) of hydrogen atom is x pm. The radius of stationary state ($n=3$) of He^+ ion (in pm) is

- A $9/8x$
- B $9x/8$
- C $16x/9$
- D $9/16x$

QUESTION 12 · TG EAPCET 2025 (Online) 2nd May Morning Shift

When electromagnetic radiation of wavelength 310 nm falls on the surface of a metal having work function 3.55 eV , the velocity of photoelectrons emitted is $x \times 10^5 \text{ ms}^{-1}$. The value of x is (Nearest integer) ($m_e = 9 \times 10^{-31} \text{ kg}$)

- A 2
- B 4
- C 5
- D 6

QUESTION 13 · TG EAPCET 2024 (Online) 11th May Morning Shift

If n, l represent the principal and azimuthal quantum numbers respectively, the formula used to know the number of radial nodes possible for a given orbital is

- A $(n-l)$
- B $(n-l+1)$
- C $(n-l-1)$
- D $(n-2)$

QUESTION 14 · TG EAPCET 2024 (Online) 11th May Morning Shift

If the radius of first orbit of hydrogen like ion is $1.763 \times 10^{-2} \text{ nm}$, the energy associated with that orbit (in J) is

- A $+1.962 \times 10^{-17}$
- B -1.962×10^{-17}
- C -0.872×10^{-17}
- D -2.18×10^{-18}

QUESTION 15 · TG EAPCET 2024 (Online) 10th May Evening Shift

The wavelength of an electron is 10^3 nm . What is its momentum in kgms^{-1} ? ($h = 6.625 \times 10^{-34} \text{ Js}$)

- A 6.625×10^{-31}
- B 6.625×10^{-37}
- C 6.625×10^{-28}
- D 6.625×10^{-34}

QUESTION 16 · TG EAPCET 2024 (Online) 10th May Evening Shift

Two statements are given below :

Statement I : In H atom, the energy of 2s and 2p orbitals is same.

Statements II : In He atom, the energy of 2s and 2p orbitals is same.

The correct answer is

- A Both statements I and II are correct.
- B Both statements I and II are not correct.
- C Statement I is correct but statement II is not correct.
- D Statement I is not correct but statement II is correct.

QUESTION 17 · TG EAPCET 2024 (Online) 10th May Morning Shift

The wavenumber of first spectral line of Lyman series of He^{+} ion is $x\text{m}^{-1}$. What is the wave number (m^{-1}) of second spectral line of Balmer series of Li^{2+} ion?

- A $(9x)/(16)$
- B $(16x)/(9)$
- C $(8x)/(27)$
- D $(27x)/(8)$

QUESTION 18 · TG EAPCET 2024 (Online) 10th May Morning Shift

The uncertainty in determination of position of a small ball of mass 10 g is 10^{-33}m . With what % of accuracy its speed can be measured, if it has a speed of 52.5 ms^{-1} ? ($h=6.6 \times 10^{-34}\text{Js}$)

- A 1.0%
- B 20%
- C 10%
- D 2.0%

QUESTION 19 · TG EAPCET 2024 (Online) 9th May Evening Shift

The kinetic energy of electrons emitted, when radiation of frequency $1.0 \times 10^{15}\text{Hz}$ hits a metal, is $2 \times 10^{-19}\text{J}$. What is the threshold frequency of the metal (in Hz)? ($h=6.6 \times 10^{-34}\text{Js}$)

- A 35×10^{15}
- B 3.3×10^{14}
- C 6.97×10^{15}
- D 6.97×10^{14}

QUESTION 20 · TG EAPCET 2024 (Online) 9th May Morning Shift

Identify the pair of species having same energy from the following. (The number given in the bracket corresponds to principal quantum number (n) in which the electron is present.)

- A $\text{H}(n=1), \text{Li}^{2+}(n=1)$
- B $\text{Li}^{2+}(n=3), \text{Be}^{3+}(n=4)$
- C $\text{He}^{+}(n=1), \text{Li}^{2+}(n=3)$
- D $\text{H}(n=3), \text{Li}^{2+}(n=3)$

QUESTION 21 · TG EAPCET 2024 (Online) 9th May Morning Shift

Which one of the following corresponds to the wavelength of line spectrum of H atom in its Balmer series ? (R = Rydberg constant)

- A $(9)/(8R)$
- B $(100)/(21R)$
- C $(25)/(24R)$
- D $(16)/(15R)$

QUESTION 22 · TS EAMCET 2023 (Online) 14th May Evening Shift

A 100 W bulb emits light of wavelength $x\text{Å}$. What is the value of x , if the number of photons emitted is $2.0 \times 10^{20}\text{s}^{-1}$?
($h=6.63 \times 10^{-34}\text{Js}$, $1\text{W}=1\text{Js}^{-1}$)

- A 3578
- B 4978
- C 3978
- D 4578

QUESTION 23 · TS EAMCET 2023 (Online) 14th May Evening Shift

The ratio of the difference in energy between the first and second Bohr orbits to that between the second and third orbit is

- A $(5)/(27)$
- B $(27)/(5)$
- C $(4)/(9)$
- D $(9)/(4)$

QUESTION 24 · TS EAMCET 2023 (Online) 14th May Morning Shift

The energy of second orbit of hydrogen atom is $-5.45 \times 10^{-19}\text{J}$. What is the energy of first orbit of Li^{2+} ions (in J)?

- A -1.962×10^{-18}
- B -1.962×10^{-17}
- C -3.924×10^{-17}
- D -3.924×10^{-18}

QUESTION 25 · TS EAMCET 2023 (Online) 14th May Morning Shift

The number of electrons with $(n+l)$ values equal to 3, 4 and 5 in an element with atomic number (Z) 24 are respectively
(n = principal quantum number and l = azimuthal quantum number)

- A 7, 8, 5
- B 6, 8, 6
- C 8, 7, 5
- D 8, 8, 5

QUESTION 26 · TS EAMCET 2023 (Online) 13th May Evening Shift

What is the approximate angular momentum (in $J s$) of electron in hydrogen atom in its ground state?
($h = 6.625 \times 10^{-34} J s$)

- A 2110×10^{-37}
- B 2110×10^{-36}
- C 1055×10^{-37}
- D 1055×10^{-36}

QUESTION 27 · TS EAMCET 2023 (Online) 13th May Evening Shift

The energy of electron in hydrogen atom when present in $n=1, n=2$ and $n=3$ will be in the ratio of

- A 25:16:9
- B 16:9:4
- C 36:9:4
- D 3:2:1

QUESTION 28 · TS EAMCET 2023 (Online) 13th May Morning Shift

The radius of first Bohr orbit of hydrogen atom is same as that of orbit (n) of hydrogen like species X. (n) and X respectively are

- A (2), Li^{2+}
- B (3), Li^{2+}
- C (2), Be^{3+}
- D (2), He^{+}

QUESTION 29 · TS EAMCET 2023 (Online) 13th May Morning Shift

Identify the impossible quantum number set for the electron from the following.

- A $n=2, l=0, m=0, s=-\frac{1}{2}$
- B $n=2, l=1, m=0, s=\frac{1}{2}$
- C $n=3, l=3, m=1, s=\frac{1}{2}$
- D $n=4, l=2, m=1, s=\frac{1}{2}$

QUESTION 30 · TS EAMCET 2023 (Online) 12th May Evening Shift

Consider the following.

- I. The electron spin quantum number describes the orientation of the spin of the nucleus with respect to the magnetic field.
- II. The orbitals represented by the quantum numbers $n=3, l=2, m=+2$ and $n=3, l=2, m=-2$ have the same energy.
- III. The energy of a photon is directly proportional to wavelength but inversely proportional to wave number.
- IV. Lyman series of lines appear in ultra-violet region.

The correct statements are

- A II and IV only
- B I and II only
- C II, III and IV only
- D I, III and IV only

Source & verification

Chapter-wise Atomic Structure PYQ archive

The examination name, year, date, and shift shown above each item identify the original paper. Mathematical notation has been normalized for clear PDF rendering.